# *International Mathematics Assessments for Schools (2023-2024)*

## Junior Division Round 2

Time: 120 minutes

*Printed Name* _____ *Code* _____ *Score* _____

## Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

**The following area is to be filled in by the judges;**
**the contestants are not supposed to mark anything here.**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | Total Score | Signature |
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| Score | | | | | | | | | | | | | | | | | |
| Score | | | | | | | | | | | | | | | | | |

# Junior Division Round 2

## Questions 1 to 5, 4 marks each

1. A car and a bus left City A at the same time and traveled along the same road to City B which was 160 km away. They ran at different but constant speeds which were integral numbers of km per hour. The difference in their speeds was a prime number of km per hour. After traveling for two hours, the distance the car had covered was five times the distance the bus still had to cover. What is the speed, in km per hour, of the bus?

（A）65　（B）66　（C）67　（D）68　（E）69

Answer：_____

2. The sum of five distinct positive integers is equal to 2024. We denote by $d$ the greatest common divisor of the 5 integers. What is the largest possible value that $d$ can take?

（A）84　（B）88　（C）92　（D）96　（E）100

Answer：_____

3. What is the number of integer solutions to the equation $1+2023x+2025y=xy$?

（A）252　（B）126　（C）63　（D）32　（E）16

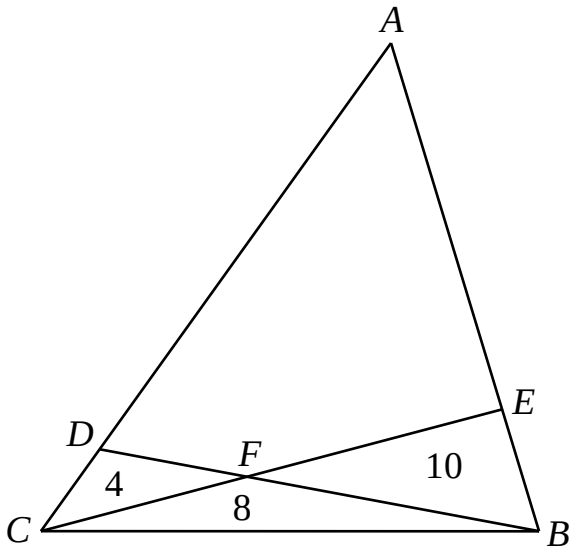
Answer：_____

4. What is the smallest positive integer $m$, such that there exists a positive integer $n$ for which the first four digits after the decimal point in the decimal representation of the fraction $\frac{n}{m}$ are 2, 0, 2, 4 in that order?

（A）89　（B）163　（C）326　（D）652　（E）1250

Answer：_____

5. In triangle *ABC*, points *D* and *E* are on the sides *AC* and *AB*, respectively, and point *F* is the intersection of *BD* and *CE*, as shown in the diagram below. If the areas of triangles *DFC*, *BCF* and *BEF* are equal to 4 $cm^2$, 8 $cm^2$ and 10 $cm^2$, respectively, what is the area, in $cm^2$, of quadrilateral *ADFE*?




（A）44　（B）46　（C）48　（D）50　（E）52

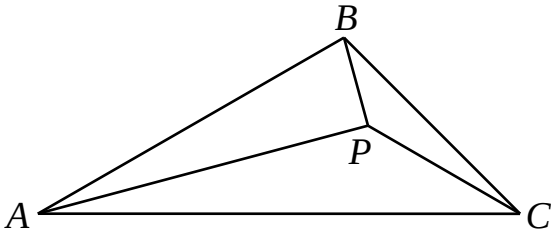
Answer：_____

## Questions 6 to 13, 5 marks each

6. It is known that $2^{84}$ is a 26-digit number so that 1 is its first digit. Among $2^1$, $2^2$, $2^3$, $2^4$, …, $2^{83}$ and $2^{84}$, how many numbers are there so that 4 is their first digit?

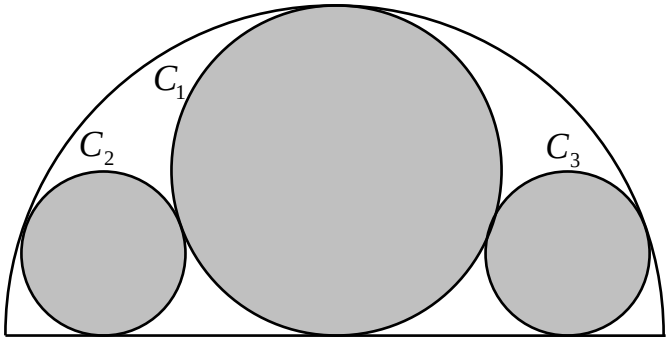
Answer：_____ numbers

7. In triangle *ABC*, $\angle BAC = 30°$ and $\angle ABC = 105°$. Let *P* be a point inside the triangle such that $\angle PAB = 15°$ and $\angle ABP = 75°$, as shown in the diagram below. What is the measure, in degrees, of the $\angle PCB$?




Answer：_____ °

8. There are one semicircle and three shaded circles, $C_1$, $C_2$ and $C_3$, where the lengths of radius of the two smaller shaded circles, $C_2$ and $C_3$, are equal. It is known that each of $C_1$, $C_2$ and $C_3$ is tangent to the semicircle and $C_1$ is tangent to each of $C_2$ and $C_3$. It is also known that each of $C_1$, $C_2$ and $C_3$ is tangent to the diameter of the semicircle, where $C_1$ and the diameter of the semicircle are tangent at the centre of the semicircle as shown in the diagram below. If the shaded area is 777 $cm^2$ in total, what is the area, in $cm^2$, of the unshaded parts of the semicircle? (Take $\pi = \frac{22}{7}$)




Answer : _____ $cm^2$

9. Consider 5 points located on the same plane, where any three of these points are not collinear. What is the maximum possible measure, in degrees, of the smallest angle among all the interior angles of a triangle with three of these points as its vertices?

Answer : _____ °

10. All positive integers with digit-sum equal to 8 are arranged in increasing order. If the $n^{th}$ number is 2024, what is the $6n^{th}$ number?

Answer : _____

11. Let $n$ be a positive integer so that the least common multiple of $n$ and 49 is a perfect square number. How many possible values of $n$ are there if $n \leq 2024$?

Answer : _____

12. There is a road from town A to town B. Along this road, there is a line of villages spaced 10 km apart between two adjacent villages, the distance between town A and the first village is 20 km and the distance between the last village and town B is 20 km. In each village, there are some statues. Arthur travels from town A to town B along this road and he must travel at least 20 km per day. If he stops at a village, then he will live in that village that day and travel to town B next day. So Arthur will miss some villages. Arthur also wants to visit statues as many as possible. For example, for a line of four villages with the following amounts of statues:

- A 4 3 4 3 B Arthur would stop at the first and third villages
- A 2 4 5 4 B Arthur would stop at the second and fourth villages
- A 5 4 2 3 B Arthur would stop at the first and fourth villages

In each case, Arthur would visit 8 statues.

In the following line of villages with the following amounts of statues:

A 3 1 4 1 5 9 2 6 5 3 5 8 9 8 7 5 2 B

what is the maximum number of statues that Arthur would visit?

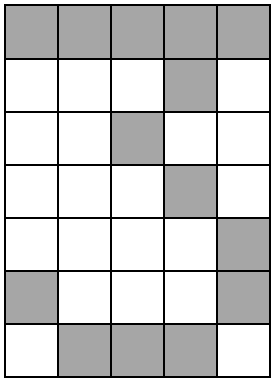
Answer： _____ statues

13. Two children, Ann and Bob, play a game in which an ideal coin is tossed repeatedly. The probability of obtaining heads or tails on each toss of a coin is equal to $\frac{1}{2}$. Ann throws the coin 4 times, while Bob throws the coin 6 times, independently of each other. Let $\frac{p}{q}$ be the probability that both children obtain the same number of heads, where the fraction is irreducible. What is the value of $p+q$?

Answer： _____

## Questions 14 to 15, 20 marks each
## (Detailed solutions are needed for these two problems)

14. Some squares of a $7 \times 5$ board are colored gray, as shown in the diagram below. What is the number of ways to choose five gray squares so that any two squares chosen are on different rows and different columns?



Answer : _____ ways

15. Triangle *ABC* is an isosceles triangle so that $AB = AC$. Let *E* be the midpoint of *AC* and *K* be the intersection of the bisector of $\angle CAB$ and the circumcircle of triangle *ABE*, as shown in the diagram below. Show that the length of diameter of the circumcircle of triangle *ABC* is equal to $\frac{4}{3}AK$.

